Jiangpeng Li

Ph.D. student in Number Theory

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Education

Ph.D. in Mathematics, Shandong University, September 2021 – March 2027 (expected). Advisor: Professor Guangshi Lü.

B.Sc. in Mathematics, Yanbian University, September 2017 – June 2021.

Research Interests

My research is in analytic number theory, with particular interests in automorphic forms and L-functions, multiplicative functions, and probabilistic models of multiplicative functions. I study the oscillation and cancellation of arithmetic functions on sparse sets, including polynomial sequences and Beatty sequences.

My work uses Dirichlet-series methods, exponential-sum estimates, second-moment methods, sieve methods, and the circle method. I also use probabilistic multiplicative models, including the Rademacher and Steinhaus models, to investigate sign changes and random-like behaviour.

Publications

1. Jiangpeng Li and Shu Luo, “Large sums of coefficients of symmetric power L-functions,” Monatshefte für Mathematik 208 (2) (2025), 325–345. DOI.
2. Jiangpeng Li, “Sum of Fourier Coefficients of Cusp Forms over Certain Polynomial,” Lithuanian Mathematical Journal. Accepted for publication.

Preprints and Manuscripts

1. R. Li and J. Li, “Odd Power Cancellation and Even Moments over Beatty Sequences” (2026+).
2. J. Li, G. S. Lü, H. Wang, and Y. C. Sun, “Short-Interval Balog–Ruzsa Bounds Across the Square-Root Scale” (2026+).
3. J. Li and G. S. Lü, “Simultaneous Shifted Square-Free and Rough Values in Short Piatetski–Shapiro Intervals” (2026+).
4. J. Li and G. S. Lü, “Weighted and Reciprocal Automorphic Coefficients in Short Intervals under Zero Localization” (2026+).
5. J. Li, “Primes Represented by $a^2 + b^4$ in Short Intervals” (2026+).
6. J. Li, “Nonvanishing of Dirichlet L-functions of Prime Conductor” (2026+).

Work in Progress

- J. Li, “Sign Changes of Rademacher Multiplicative Functions over Sums of Two Squares.”

Talks

- 2026 Shandong University Number Theory Conference, plenary speaker, July 9, 2026.

Teaching and Academic Service

- Teaching Assistant, Shandong University. Assisted with the courses Algebraic Geometry and Mathematical Analysis, including problem sessions, assignment grading, and student consultation.
- Graduate Academic Affairs Intern, Shandong University. Assisted with course administration, teaching support, and the organization and maintenance of academic records.